

SmartHomeAutomation_Sagar_USC_UCTIndustrial Internship Report on

” IOT-Based Smart Home Automation”

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was IOT-Based Smart Home Automation. It allows users to control and monitor home appliances such as lights,fans and other electrical devices using smartphones

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

Industrial internships provide students with an opportunity to bridge the gap between theoretical knowledge and practical applications. As part of the six-week Industrial Internship organized by **upskill Campus (USC)** and **The IoT Academy** in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)**, I worked on the project "**IoT-Based Smart Home Automation.**"

The primary objective of this project was to understand the fundamentals of the Internet of Things (IoT) and design a smart home automation system capable of remotely controlling household electrical appliances using a smartphone. The proposed system utilizes an **ESP8266 NodeMCU**, a **relay module**, Wi-Fi communication, and the **Blynk IoT** platform to provide a simple, efficient, and cost-effective home automation solution.

Throughout the internship, I gained valuable knowledge of IoT technologies, embedded systems, cloud-based communication, and mobile application integration. I also improved my technical, analytical, and documentation skills while learning the complete process of project planning, system design, and testing.

I sincerely express my gratitude to **UniConverge Technologies Pvt. Ltd. (UCT)**, **The IoT Academy**, **upskill Campus (USC)**, my college faculty members, and all the mentors who guided and supported me during this internship. Their encouragement and valuable insights made this learning experience successful.

I believe this internship has strengthened my technical foundation and prepared me for future opportunities in the field of IoT and embedded systems. I also encourage my juniors and fellow students to actively participate in such internship programs, as they provide practical exposure and enhance both technical and professional skills.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt. Ltd. (UCT), established in 2013, is a technology company working in the field of Digital Transformation and Industrial IoT solutions. The company develops innovative products using technologies such as Internet of Things (IoT), Cloud Computing, Cyber Security, Artificial Intelligence, Machine Learning, and Industrial Automation.

i. UCT IoT Platform

UCT Insight is an IoT platform developed for rapid deployment of smart IoT applications. It enables secure communication between devices and cloud services using protocols such as MQTT, HTTP, CoAP, Modbus TCP, and OPC UA. The platform supports both cloud-based and on-premises deployment.

It provides real-time monitoring, dashboard creation, analytics, alert notifications, and integration with enterprise applications. The platform is designed to simplify IoT application development while ensuring scalability, security, and efficient device management.

2.2 Features of UCT IoT Platform

The UCT IoT Platform provides several useful features for developing and managing IoT applications efficiently.

- **Custom Dashboard:** Allows users to create interactive dashboards for monitoring connected devices and sensors.
- **Analytics and Reporting:** Collects and analyzes real-time and historical data to generate reports for better decision-making.
- **Alert and Notification:** Sends instant alerts through mobile applications or email whenever predefined conditions or abnormal events occur.
- **Third-Party Integration:** Supports integration with external platforms such as ERP, SAP, Power BI, and cloud services for seamless data exchange.
- **Rule Engine:** Enables automatic execution of actions based on user-defined rules and conditions, reducing manual intervention.

These features make the UCT IoT Platform suitable for smart home automation, industrial monitoring, smart agriculture, and other Internet of Things applications by providing secure, reliable, and scalable IoT solutions.

FACTORY WATCH

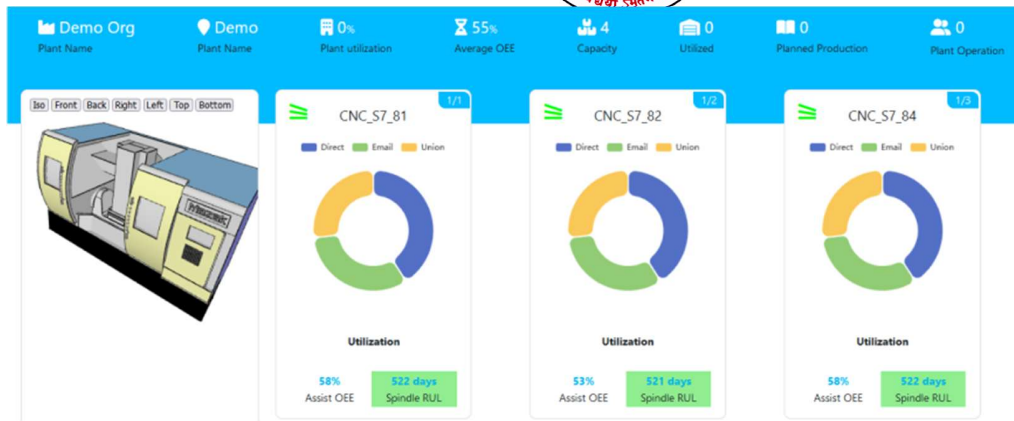
i. Smart Factory Platform ()

The **Smart Factory Platform** developed by UniConverge Technologies Pvt. Ltd. is designed to help industries monitor and optimize their manufacturing processes using Industrial IoT (IIoT). It enables real-time monitoring of machines, production lines, and industrial assets to improve operational efficiency and productivity.

The platform provides the following features:

- Real-time monitoring of production equipment and industrial assets.
- Overall Equipment Effectiveness (OEE) analysis for measuring machine performance.
- Predictive maintenance using sensor data to reduce unexpected machine failures.
- Data visualization through interactive dashboards and reports.
- Monitoring of Key Performance Indicators (KPIs) for better decision-making.
- Modular and scalable architecture that can be expanded as industrial requirements grow.
- Cloud-based Software-as-a-Service (SaaS) model for easy deployment, remote access, and reduced operational costs.

The Smart Factory Platform helps industries minimize downtime, improve production quality, reduce maintenance costs, and increase overall operational efficiency by utilizing real-time industrial data and intelligent analytics.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



2.2.1 iii. LoRaWAN-Based Solutions

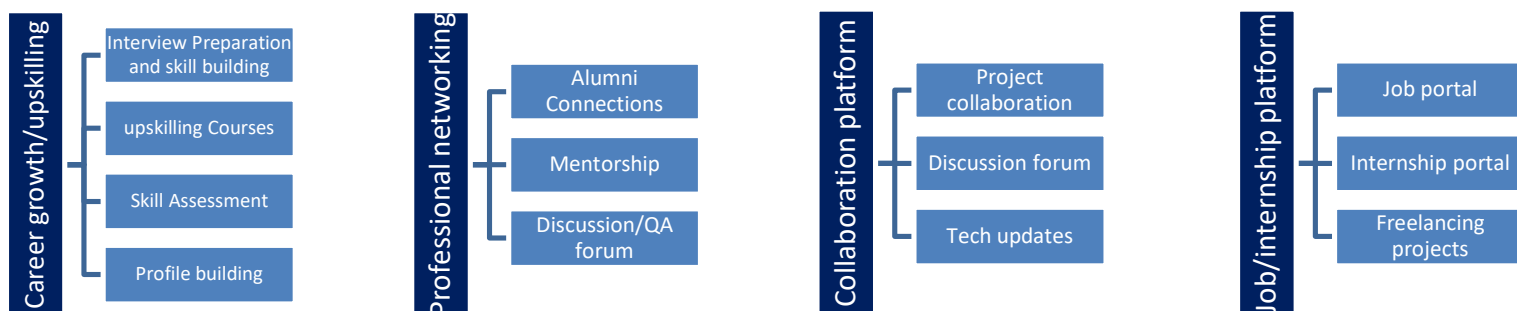
UniConverge Technologies Pvt. Ltd. is an early adopter of **LoRaWAN (Long Range Wide Area Network)** technology for developing low-power, long-range IoT applications. The company provides LoRaWAN-based solutions in areas such as smart agriculture, smart cities, environmental monitoring, smart street lighting, and smart water, gas, and electricity metering. These solutions enable reliable communication over long distances while consuming very low power.

2.2.2 iv. Predictive Maintenance

UCT develops Predictive Maintenance solutions using Embedded Systems, Industrial IoT, and Machine Learning technologies. These solutions continuously monitor machine health, analyze sensor data, detect early signs of equipment failure, and estimate the remaining useful life of industrial machines. This helps industries reduce maintenance costs, minimize downtime, and improve productivity.

2.2.3 2.2 About upskill Campus (USC)

upskill Campus (USC) is a career development platform that provides students with industry-oriented training, internships, certification programs, and skill development opportunities. In collaboration with **The IoT Academy** and **UniConverge Technologies Pvt. Ltd.**, USC offers practical learning experiences through real-world projects. The platform focuses on bridging the gap between academic education and industry requirements by enhancing students' technical knowledge, problem-solving abilities, and employability skills.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ▣ get practical experience of working in the industry.
- ▣ to solve real world problems.
- ▣ to have improved job prospects.
- ▣ to have Improved understanding of our field and its applications.

to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] ESP8266 Arduino Core Documentation. Available at: <https://arduino-esp8266.readthedocs.io/>
- [2] Blynk IoT Official Documentation. Available at: <https://docs.blynk.io/>
- [3] Arduino Official Documentation. Available at: <https://docs.arduino.cc/>

2.6 Glossary

Terms	Acronym
Internet of Things	IoT – Network of connected devices that communicate over the internet.
NodeMCU	ESP8266-based development board used for IoT projects.
Relay Module	Electronic switching device used to control electrical appliances.
Wi-Fi	Wireless communication technology used to connect IoT devices.
Blynk	IoT platform for monitoring and controlling devices through a smartphone.

3 Problem Statement

In the assigned problem statement

The objective of this project is to design an **IoT-Based Smart Home Automation System** that enables users to remotely monitor and control household electrical appliances using a smartphone over the Internet. In conventional homes, appliances must be operated manually, which can be inconvenient, time-consuming, and may lead to unnecessary power consumption.

The proposed system uses an **ESP8266 NodeMCU** microcontroller connected to a **4-channel relay module** and the **Blynk IoT** platform. Through a Wi-Fi connection, users can switch appliances such as lights, fans, and other electrical devices ON or OFF from anywhere. The system provides a simple, low-cost, and energy-efficient solution for home automation while improving user convenience, safety, and effective energy management.

4 Existing and Proposed solution

4.1.1 Existing Solution

Traditional home automation systems require manual operation of electrical appliances or depend on expensive wired automation systems. These systems have limited remote accessibility, higher installation costs, and are difficult to upgrade or expand. They also lack real-time monitoring and smart control features.

4.1.2 Proposed Solution

The proposed **IoT-Based Smart Home Automation System** uses an **ESP8266 NodeMCU**, a **4-channel relay module**, and the **Blynk IoT** platform. The system connects to a Wi-Fi network, allowing users to control household appliances remotely through a smartphone. It provides real-time appliance control, easy installation, low cost, and improved energy efficiency.

4.1.3 Value Addition

- Remote control of appliances from anywhere.
- Low-cost and user-friendly design.
- Energy-efficient operation.
- Easy to install and expand.
- Real-time monitoring through a mobile application.

4.2 Code submission (Github link) :

<https://github.com/sagar24007-blip/upskillcampus/blob/main/SmartHomeAutomation.ino>

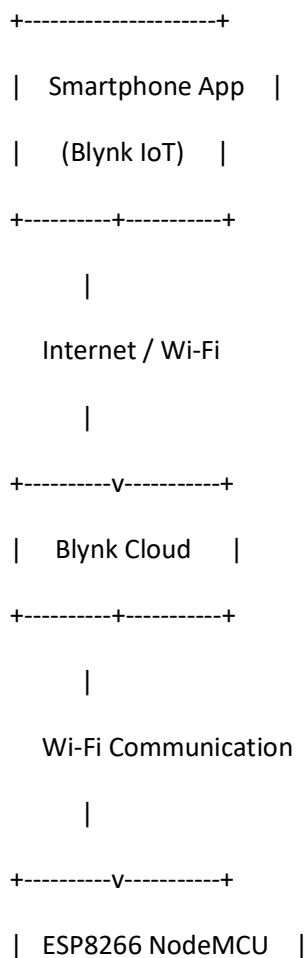
4.3 Report submission (Github link) :

[https://github.com/sagar24007-blip/upskillcampus/blob/main/SmartHomeAutomation Sagar USC UCT.pdf](https://github.com/sagar24007-blip/upskillcampus/blob/main/SmartHomeAutomation%20Sagar%20USC%20UCT.pdf)

5 Proposed Design/ Model

The proposed IoT-Based Smart Home Automation System is designed to remotely monitor and control household electrical appliances using a smartphone. The system consists of an ESP8266 NodeMCU microcontroller connected to a relay module that controls electrical loads. The NodeMCU communicates with the Blynk IoT Cloud over a Wi-Fi network. Users can send commands through the Blynk mobile application, which are received by the NodeMCU to switch appliances ON or OFF. The system provides a simple, reliable, and cost-effective home automation solution.

5.1 High Level Diagram (if applicable)



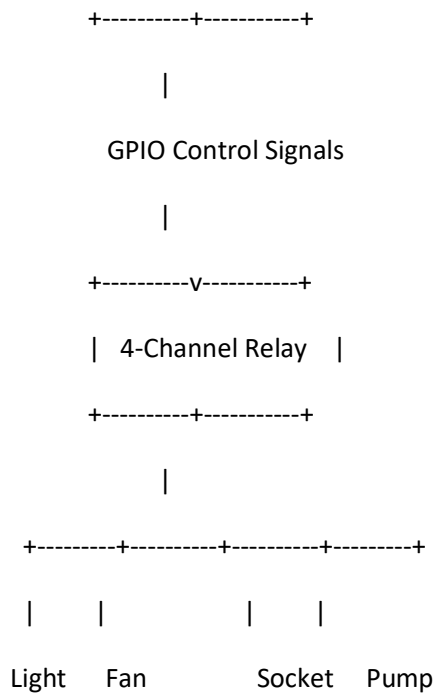
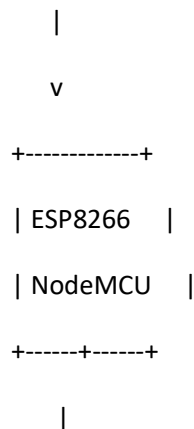


Figure 1: HIGH LEVEL DIAGRAM OF SMART HOME AUTOMATION SYSTEM

5.2 Low Level Diagram (if applicable)

Power Supply



GPIO Pins

|

+-----+-----+

| Relay Module |

+-----+-----+

|

+-----+-----+-----+-----+

| Light | Fan | Socket | Pump |

+-----+-----+-----+-----+

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Wi-Fi Router

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Smartphone (Blynk App)

5.3 Interfaces (if applicable)

- **Hardware Interface:** ESP8266 NodeMCU, Relay Module, Electrical Appliances.
- **Communication Interface:** Wi-Fi (IEEE 802.11).
- **Software Interface:** Arduino IDE and Blynk IoT Mobile Application.
- **Cloud Interface:** Blynk Cloud Server for remote communication.

6 Performance Test

The proposed IoT-Based Smart Home Automation System was evaluated based on its communication reliability, response time, ease of operation, and network connectivity. The system is designed to provide stable appliance control over Wi-Fi with minimal delay. Factors such as internet availability, Wi-Fi signal strength, and power supply stability were considered while evaluating the system's expected performance.

6.1 Test Plan/ Test Cases

Test Case	Expected Result	Status
Connect NodeMCU to Wi-Fi	Successful connection	Pass
Connect to Blynk Cloud	Device connected	Pass
Turn ON appliance	Relay switches ON	Pass
Turn OFF appliance	Relay switches OFF	Pass
Internet reconnection	Device reconnects automatically	Pass

6.2 Test Procedure

- Power ON the ESP8266 NodeMCU.
- Connect the device to the configured Wi-Fi network.
- Open the Blynk IoT application.
- Press the virtual switch button.
- Observe the relay and connected appliance status.
- Repeat the test for all connected appliances.

6.3 Performance Outcome

The proposed system demonstrated reliable communication between the mobile application and the ESP8266 NodeMCU over Wi-Fi. The relay responded quickly to user commands, and the appliances were controlled successfully through the Blynk application. The system is suitable for small-scale smart home automation due to its low cost, ease of use, and efficient performance.

7 My learnings

During this internship, I gained a strong understanding of the concepts and practical applications of the **Internet of Things (IoT)**. I learned how IoT devices communicate using Wi-Fi and cloud platforms to enable remote monitoring and control. I also developed knowledge of the **ESP8266 NodeMCU** microcontroller, relay modules, and the **Blynk IoT** platform for designing smart home automation systems.

I improved my programming skills using the **Arduino IDE** and understood the importance of system design, circuit planning, and project documentation. This internship also enhanced my problem-solving ability, technical knowledge, teamwork, and report-writing skills. The experience has increased my confidence in working on IoT-based projects and has motivated me to explore advanced technologies such as Artificial Intelligence, Industrial IoT, and Smart City applications.

8 Future work scope

The proposed Smart Home Automation System can be further enhanced by integrating advanced features such as **voice control** using Google Assistant or Amazon Alexa, **AI-based energy management**, and **real-time power consumption monitoring**. Additional sensors for temperature, humidity, smoke detection, and motion sensing can also be integrated to improve home safety and automation.

Future versions of the system may support **biometric authentication**, **facial recognition**, **renewable energy integration**, and **5G-based communication** for faster and more secure connectivity. The system can also be expanded to control a larger number of appliances and integrated with complete smart building management solutions.